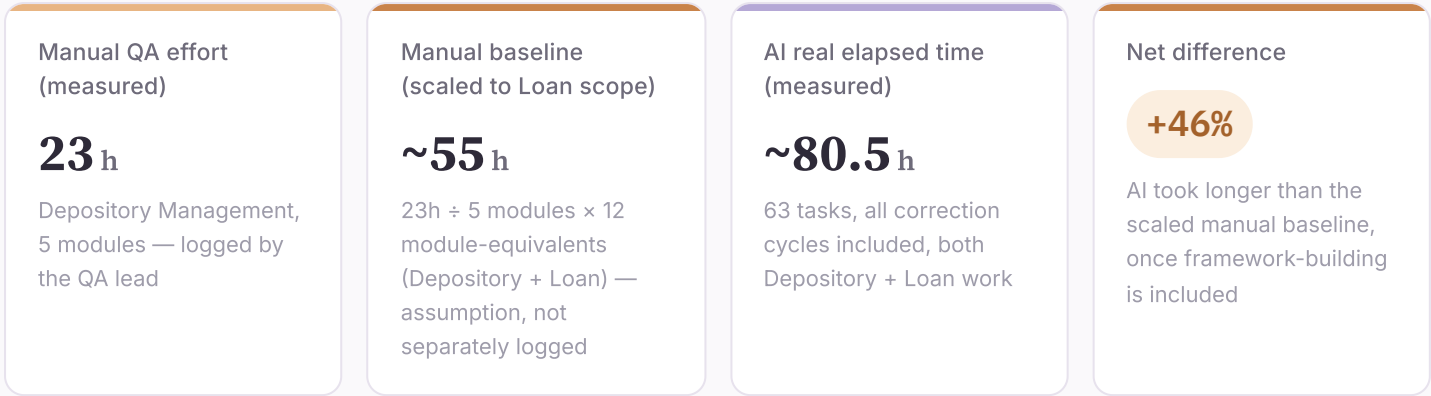
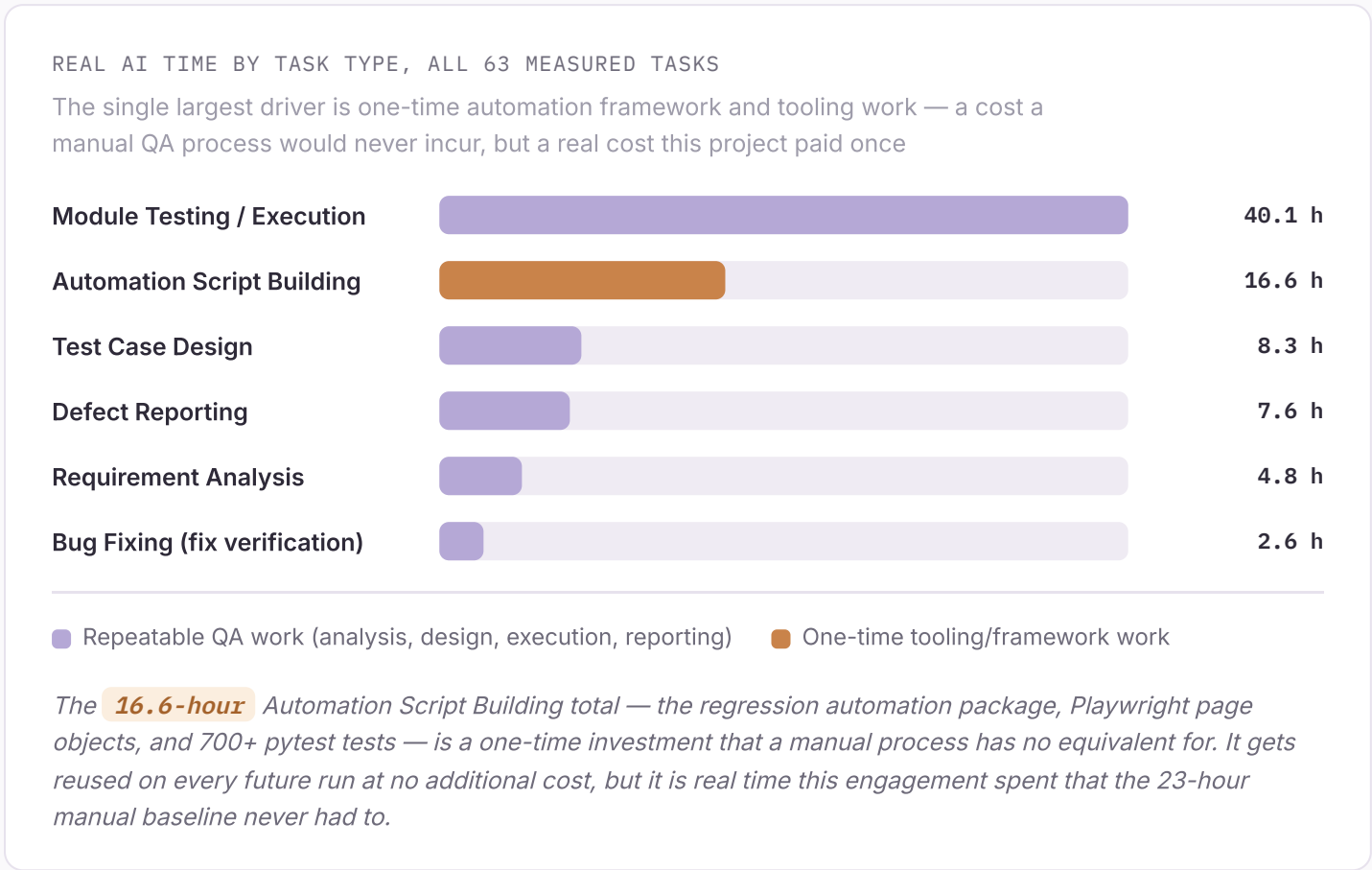


AI-Augmented Quality Assurance for Sage Intacct

An honest look at how long AI-assisted QA actually took — measured from real session timestamps, including exploration, mistakes, and correction cycles, not just the time a clean final result took to produce.



Why AI Took Longer Overall



Time — Requirement Analysis & Test Case Design

MANUAL VS. AI-ASSISTED, DEPOSITORY ACCOUNT (THE ONE MODULE WITH A DIRECTLY-LOGGED MANUAL COMPARISON)

The 23h manual figure covers analysis + design + execution + reporting across all 5 Depository modules combined, not broken out by phase or module — so it cannot be split cleanly against a single module's AI time. The comparison below uses only what both sides actually have in common.

Requirement Analysis + Test Case Design, combined

real AI time: 42 min

AI

42 min

AI-assisted (real elapsed, Depository Account, this is the only module with both a manual reference and a complete AI pass)

No manual time was separately logged for analysis and design on Depository Account alone, so no percentage is shown here — only the real AI figure. The 5h/6h manual split shown elsewhere in this report is a labeled planning estimate for the whole 5-module Depository effort, not a per-module measurement, and should not be divided against a single module's AI time.

Execution Volume — Cases Run per Module

AI-ASSISTED EXECUTION, LENDING & DEPOSITORY MANAGEMENT

Test cases executed and results recorded (pass/fail + evidence) across the project

Loan Account

177 cases

Interest Rate

100 cases

Category

65 cases

Fee Type

50 cases

Cases executed + result documented

392 cases across these 4 modules → 90 defect reports drafted from the failures, each pre-formatted and cross-referenced to its test case before triage. This volume took ~40.1 real hours of Module Testing / Execution time across the whole project (21 separate testing sessions, 10–519 minutes each) — not a single clean pass.

Phase-by-Phase Comparison

Manual = QA lead's real log, Depository scope (23h, the only directly-logged manual figure). AI = real elapsed time measured from session transcripts across the whole project, task type by task type — includes every correction cycle, not clean execution only.

Task Type	Tasks	Real AI Time	Average	Range	What Drove the Time
Requirement Analysis	7	4.8 h	41 min	5 – 79 min	Ranges from a quick ticket lookup to a full environment/credential switch requiring several corrections
Test Case Design	7	8.3 h	71 min	4 – 230 min	Most tasks were quick; one (Lending Workbench) spanned multiple rounds of scope correction
Module Testing / Execution	21	40.1 h	115 min	10 – 519 min	The largest category by far; includes environment outages and forced logouts counted as real elapsed time
Defect Reporting	8	7.6 h	57 min	1 – 215 min	Several rounds of bug-report reformatting to match house style added real time here
Automation Script Building	16	16.6 h	62 min	1 – 215 min	One-time framework/tooling cost — not part of a repeatable QA cycle, but real time spent
Bug Fixing (fix verification)	3	2.6 h	52 min	11 – 120 min	Retesting dev-reported fixes against the live app

Overall Impact

The only module with both a real manual reference point and a complete AI-assisted pass (analysis, design, execution, and defect reporting) is Depository Account. Its real AI time was **42 minutes** for analysis and design combined; no separate manual figure exists for that module alone to compare it against, so no percentage is claimed here — only the measured AI number.

Scaled to cover the whole project (Depository + Loan Management, using the Depository 23h/5-module rate as an assumed per-module baseline), real AI time across all task types was **~80.5 hours against a ~55-hour scaled manual estimate – about 46% longer**, driven mainly by one-time automation framework and tooling work with no manual equivalent. Strip that one-time cost out and the remaining repeatable QA work (analysis, design, execution, reporting) totals **~63.9 hours** against the same ~55-hour scaled baseline — still somewhat longer, not faster, once real correction cycles are counted.

This is AI-*augmented* QA, not automated QA, and this report is not claiming a blanket speed-up. What it shows is: AI-assisted analysis and design work is fast in absolute terms (minutes, not hours) even including corrections; execution and reporting throughput is high (392 cases, 90 defect reports); and building the automation capability itself was a real, one-time cost that pays off on future runs but should not be hidden from a first-engagement comparison.

Risks & Limitations

- The 55-hour "scaled" manual baseline for the combined Depository + Loan scope is an assumption (23h ÷ 5 modules, applied evenly to 12 module-equivalents) — no manual time was ever separately logged for the Loan modules, so this number should be read as an estimate, not a measurement.
- AI output still needs human review before it's final — ambiguous requirements or missing business context can be misread, and several of the real hours above are Claude correcting its own mistakes after review.

- Live execution depends on environment stability — this engagement hit a broken deploy and repeated concurrent-session conflicts that inflated real elapsed time and would have cost a manual tester time too, though not identically.
- One-time framework-building cost (16.6h) will not recur on future modules at the same rate — a fair multi-cycle comparison would need to track whether execution and reporting stay fast on runs 2, 3, and beyond, once the framework is already built.

Sources: LME_Regression_Consolidated.xlsx (Jira Bugs sheet · Loan Account/Interest Rate/Category/Fee Type CRUD sheets) · Depository_Account_CRUD_Report.html · TEST_CASES_Depository_Account.xlsx · CHECKLIST_Depository_Management_Regression.xlsx · real elapsed-time analysis of 9 session transcripts (2026-07-15 to 2026-08-18), 63 tasks measured.